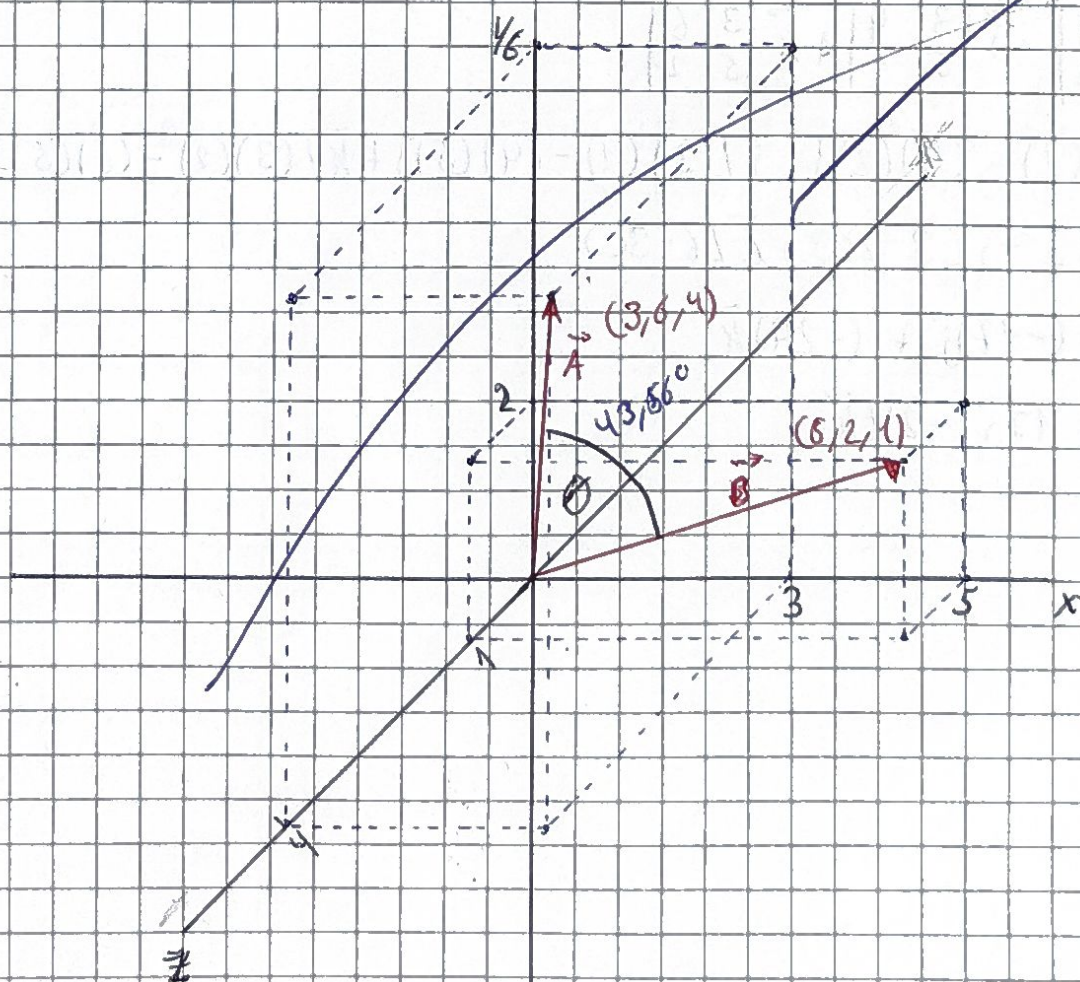


Universidad Central del Ecuador
Facultad de Filosofía, Letras, Ciencias de la Educación
Pedagogía de las Ciencias Experimentales e Informáticas

Nombre: Juan Valle
Curso: 5^{to} "B"
Fecha: 13/05/2026

Vectores

1. $\vec{A} = 3\vec{i} + 6\vec{j} + 4\vec{k}$
 $\vec{B} = 5\vec{i} + 2\vec{j} + \vec{k}$



Modulo de $\vec{A}$
 $|\vec{A}| = \sqrt{(3)^2 + (6)^2 + (4)^2}$
 $|\vec{A}| = \sqrt{9 + 36 + 16}$
 $|\vec{A}| = \sqrt{61}$
 $|\vec{A}| \approx 7,81$

Modulo de $\vec{B}$
 $|\vec{B}| = \sqrt{(5)^2 + (2)^2 + (1)^2}$
 $|\vec{B}| = \sqrt{25 + 4 + 1}$
 $|\vec{B}| = \sqrt{30}$
 $|\vec{B}| \approx 5,48$

Producto Escalar ($\vec{A} \cdot \vec{B}$)

$$\vec{A} \cdot \vec{B} = (A_x B_x) + (A_y B_y) + (A_z B_z)$$

$$\vec{A} \cdot \vec{B} = (3)(5) + (6)(2) + (4)(1)$$

$$\vec{A} \cdot \vec{B} = 15 + 12 + 4$$

$$\vec{A} \cdot \vec{B} = 31$$

Módulo del producto vectorial

$$|\vec{A} \times \vec{B}| = |\vec{A}| \cdot |\vec{B}| \cdot \sin(\theta)$$

$$|\vec{A} \times \vec{B}| = (7.81)(5.48) \cdot \sin(43.56^\circ)$$

$$|\vec{A} \times \vec{B}| = (7.81)(5.48) \cdot 0.689$$

$$|\vec{A} \times \vec{B}| \approx 29.48$$

Ángulo entre los vectores (θ)

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|}$$

$$\cos \theta = \frac{31}{\sqrt{61} \cdot \sqrt{30}}$$

$$\cos \theta = \frac{31}{\sqrt{1830}}$$

$$\cos \theta = \frac{31}{42.78}$$

$$\cos \theta \approx 0.7246$$

$$\theta = \cos^{-1}(0.7246)$$

$$\theta \approx 43.56^\circ$$

Producto Vectorial

$$\vec{C} = \vec{i} \begin{vmatrix} 6 & 4 \\ 2 & 1 \end{vmatrix} - \vec{j} \begin{vmatrix} 3 & 4 \\ 5 & 1 \end{vmatrix} + \vec{k} \begin{vmatrix} 3 & 6 \\ 5 & 2 \end{vmatrix}$$

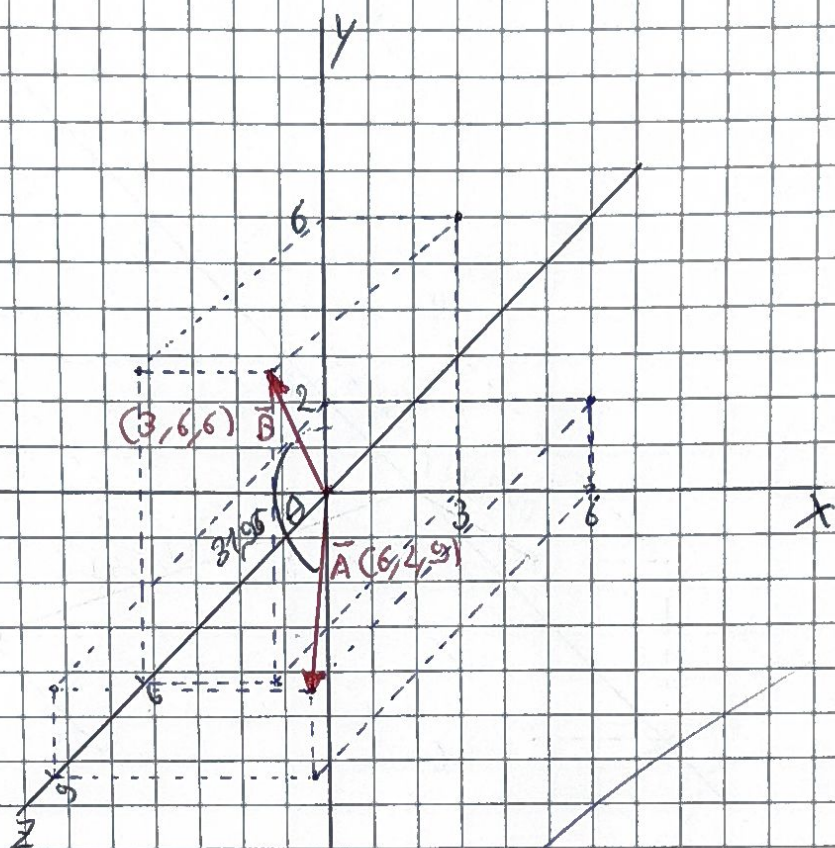
$$\vec{C} = \vec{i} [(6)(1) - (4)(2)] - \vec{j} [(3)(1) - (4)(5)] + \vec{k} [(3)(2) - (6)(5)]$$

$$\vec{C} = \vec{i} [6 - 8] - \vec{j} [3 - 20] + \vec{k} [6 - 30]$$

$$\vec{C} = -2\vec{i} - (-17)\vec{j} + (-24)\vec{k}$$

$$\vec{C} = -2\vec{i} + 17\vec{j} - 24\vec{k}$$

$$\begin{aligned} \vec{A} &= 6\vec{i} + 2\vec{j} + 9\vec{k} \\ \vec{B} &= 3\vec{i} + 6\vec{j} + 6\vec{k} \end{aligned}$$



Modulo de A

$$\begin{aligned} |\vec{A}| &= \sqrt{(6)^2 + (2)^2 + (9)^2} \\ |\vec{A}| &= \sqrt{36 + 4 + 81} \\ |\vec{A}| &= \sqrt{121} \\ |\vec{A}| &= 11 \end{aligned}$$

Modulo de B

$$\begin{aligned} |\vec{B}| &= \sqrt{(3)^2 + (6)^2 + (6)^2} \\ |\vec{B}| &= \sqrt{9 + 36 + 36} \\ |\vec{B}| &= \sqrt{81} \\ |\vec{B}| &= 9 \end{aligned}$$

Producto Escalar ($\vec{A} \cdot \vec{B}$)

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x B_x) + (A_y B_y) + (A_z B_z) \\ \vec{A} \cdot \vec{B} &= (6)(3) + (2)(6) + (9)(6) \\ \vec{A} \cdot \vec{B} &= 18 + 12 + 54 \\ \vec{A} \cdot \vec{B} &= 84 \end{aligned}$$

Angulo entre los vectores (θ)

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

$$\cos \theta = \frac{84}{(11)(9)}$$

$$\cos \theta = \frac{84}{99}$$

$$\cos \theta = 0,8485$$

$$\theta = \cos^{-1}(0,8485)$$

$$\theta \approx 31,95^\circ$$

Area del paralelogramo

$$|\vec{A} \times \vec{B}| = |\vec{A}| \cdot |\vec{B}| \cdot \sin(\theta)$$

$$|\vec{A} \times \vec{B}| = 11 \cdot 9 \cdot \sin(31,95^\circ)$$

$$|\vec{A} \times \vec{B}| = 99 \cdot 0,5292$$

$$|\vec{A} \times \vec{B}| = 52,39$$

Producto Vectorial ($\vec{A} \times \vec{B}$)

$$C = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 2 & 9 \\ 3 & 6 & 6 \end{vmatrix}$$

$$\vec{C} = \vec{i} \begin{vmatrix} 2 & 9 \\ 6 & 6 \end{vmatrix} - \vec{j} \begin{vmatrix} 6 & 9 \\ 3 & 6 \end{vmatrix} + \vec{k} \begin{vmatrix} 6 & 2 \\ 3 & 6 \end{vmatrix}$$

$$\vec{C} = -42\vec{i} - 9\vec{j} + 30\vec{k}$$

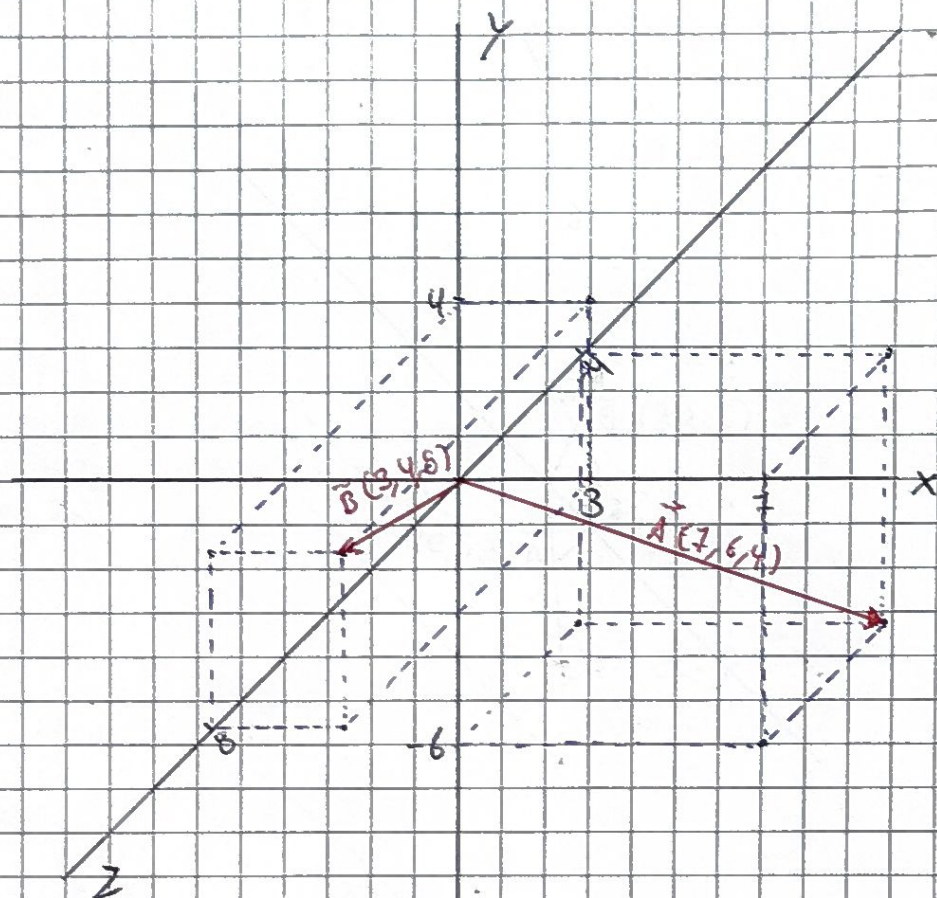
$$\vec{C} = \vec{i}[(2)(6) - (9)(6)] - \vec{j}[(6)(6) - (9)(3)] + \vec{k}[(6)(6) - (2)(3)]$$

$$\vec{C} = \vec{i}[12 - 54] - \vec{j}[36 - 27] + \vec{k}[36 - 6]$$

$$\vec{C} = -42\vec{i} - (9)\vec{j} + (30)\vec{k}$$

$$\vec{A} = 7\vec{i} - 6\vec{j} - 4\vec{k}$$

$$\vec{B} = 3\vec{i} + 4\vec{j} + 8\vec{k}$$



Modulo de A

$$|\vec{A}| = \sqrt{(7)^2 + (-6)^2 + (-4)^2}$$

$$|\vec{A}| = \sqrt{49 + 36 + 16}$$

$$|\vec{A}| = \sqrt{101}$$

$$|\vec{A}| \approx 10,05$$

Modulo de B

$$|\vec{B}| = \sqrt{(3)^2 + (4)^2 + (8)^2}$$

$$|\vec{B}| = \sqrt{9 + 16 + 64}$$

$$|\vec{B}| = \sqrt{89}$$

$$|\vec{B}| \approx 9,43$$

Producto Escalar ($\vec{A} \cdot \vec{B}$)

$$\vec{A} \cdot \vec{B} = (A_x B_x) + (A_y B_y) + (A_z B_z)$$

$$\vec{A} \cdot \vec{B} = (7)(3) + (-6)(4) + (-4)(8)$$

$$\vec{A} \cdot \vec{B} = 21 - 24 - 32$$

$$\vec{A} \cdot \vec{B} = -35$$

Angulo entre los Vectores

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

$$\cos \theta = \frac{-35}{\sqrt{101} \cdot \sqrt{89}}$$

$$\cos \theta = \frac{-35}{\sqrt{9989}}$$

$$\cos \theta = \frac{-35}{94,81}$$

$$\cos \theta \approx -0,3692$$

$$\theta = \cos^{-1}(-0,3692)$$

$$\theta \approx 111,66^\circ$$

Producto Vectorial ($\vec{A} \times \vec{B}$)

$$\vec{C} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 7 & -6 & -4 \\ 3 & 4 & 8 \end{vmatrix}$$

$$\vec{C} = \vec{i} [(-6)(8) - (-4)(4)] - \vec{j} [(7)(8) - (-4)(3)] + \vec{k} [(7)(4) - (-6)(3)]$$

$$\vec{C} = \vec{i} [-48 - (-16)] - \vec{j} [56 - (-12)] + \vec{k} [28 - (-18)]$$

$$\vec{C} = \vec{i} [-48 + 16] - \vec{j} [56 + 12] + \vec{k} [28 + 18]$$

$$\vec{C} = -32\vec{i} - 68\vec{j} + 46\vec{k}$$

Area del paralelogramo:

$$|\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin(\theta)$$

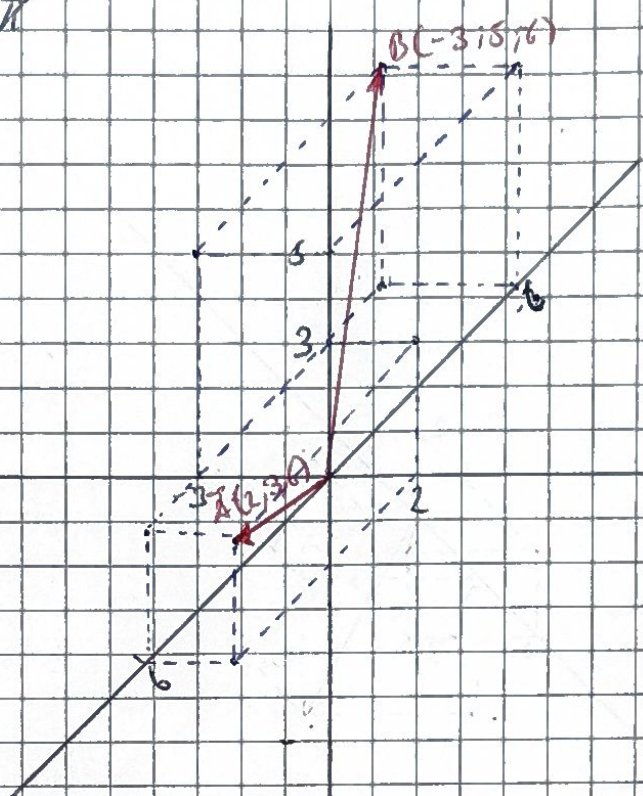
$$|\vec{A} \times \vec{B}| = 10,05 \cdot 9,43 \cdot \sin(111,66^\circ)$$

$$|\vec{A} \times \vec{B}| = 94,77 \cdot 0,9294$$

$$|\vec{A} \times \vec{B}| \approx 88,11$$

$$4. \vec{A} = 2\vec{i} + 3\vec{j} + 6\vec{k}$$

$$\vec{B} = -3\vec{i} + 5\vec{j} - 6\vec{k}$$



Modulo de A

$$|\vec{A}| = \sqrt{(2)^2 + (3)^2 + (6)^2}$$

$$|\vec{A}| = \sqrt{4 + 9 + 36}$$

$$|\vec{A}| = \sqrt{49}$$

$$|\vec{A}| = 7$$

Modulo de B

$$|\vec{B}| = \sqrt{(-3)^2 + (5)^2 + (-6)^2}$$

$$|\vec{B}| = \sqrt{9 + 25 + 36}$$

$$|\vec{B}| = \sqrt{70}$$

$$|\vec{B}| = 8,37$$

Producto escalar

$$\vec{A} \cdot \vec{B} = (A_x B_x) + (A_y B_y) + (A_z B_z)$$

$$\vec{A} \cdot \vec{B} = (2)(-3) + (3)(5) + (6)(-6)$$

$$\vec{A} \cdot \vec{B} = -6 + 15 - 36$$

$$\vec{A} \cdot \vec{B} = -27$$

Angulo entre las vectores (θ)

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

$$\cos \theta = \frac{-27}{7 \cdot \sqrt{70}}$$

$$\cos \theta = \frac{-27}{58,06}$$

$$\cos \theta = -0,461$$

Producto Vectorial

$$\vec{C} = \vec{i} \begin{vmatrix} 3 & 6 \\ 5 & -6 \end{vmatrix} - \vec{j} \begin{vmatrix} 2 & 6 \\ -3 & -6 \end{vmatrix} + \vec{k} \begin{vmatrix} 2 & 3 \\ -3 & 5 \end{vmatrix}$$

$$\vec{C} = \vec{i}[(3)(-6) - (6)(5)] - \vec{j}[(2)(-6) - (6)(-3)] + \vec{k}[(2)(5) - (-3)(3)]$$

$$\vec{C} = \vec{i}[-48] - \vec{j}[-12 + 18] + \vec{k}[10 + 9]$$

$$\vec{C} = -48\vec{i} - 6\vec{j} + 19\vec{k}$$

Area del paralelogramo

$$|\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin(\theta)$$

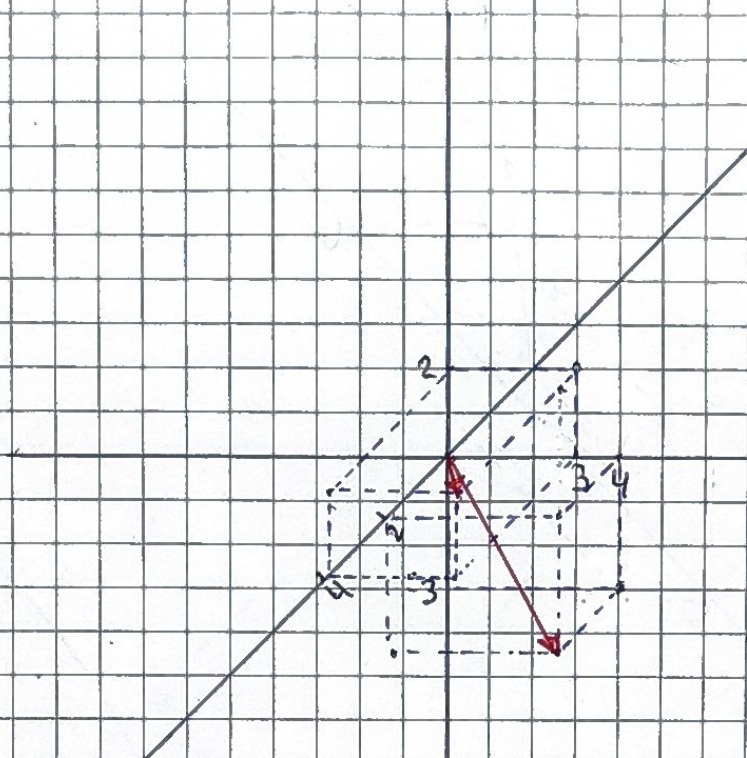
$$|\vec{A} \times \vec{B}| = 7 \cdot 8,366 \cdot \sin(117,45^\circ)$$

$$|\vec{A} \times \vec{B}| = 58,06 \cdot 0,8874$$

$$|\vec{A} \times \vec{B}| = 51,97$$

$$5. \vec{A} = 3\vec{i} + 2\vec{j} + 4\vec{k}$$

$$\vec{B} = 4\vec{i} - 3\vec{j} + 2\vec{k}$$



Modulo de A

$$|\vec{A}| = \sqrt{(3)^2 + (2)^2 + (4)^2}$$

$$|\vec{A}| = \sqrt{9 + 4 + 16}$$

$$|\vec{A}| = \sqrt{29}$$

$$|\vec{A}| = \sqrt{29}$$

$$|\vec{A}| = 5,39$$

Modulo de B

$$|\vec{B}| = \sqrt{(4)^2 + (-3)^2 + (2)^2}$$

$$|\vec{B}| = \sqrt{16 + 9 + 4}$$

$$|\vec{B}| = \sqrt{29}$$

$$|\vec{B}| = 5,39$$

Producto Escalar

$$\vec{A} \cdot \vec{B} = (A_x B_x) + (A_y B_y) + (A_z B_z)$$

$$\vec{A} \cdot \vec{B} = (3)(4) + (2)(-3) + (4)(2)$$

$$\vec{A} \cdot \vec{B} = 12 - 6 + 8$$

$$\vec{A} \cdot \vec{B} = 14$$

Angulo de los Vectores

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

$$\cos \theta = \frac{14}{\sqrt{29} \cdot \sqrt{29}}$$

$$\cos \theta = \frac{14}{29}$$

$$\cos \theta = 0,4827$$

$$\theta = \cos^{-1}(0,4827)$$

$$\theta = 61,13^\circ$$

Producto Vectorial

$$\vec{C} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & 4 \\ 4 & -3 & 2 \end{vmatrix}$$

$$\vec{C} = \vec{i} \begin{vmatrix} 2 & 4 \\ -3 & 2 \end{vmatrix} - \vec{j} \begin{vmatrix} 3 & 4 \\ 4 & 2 \end{vmatrix} + \vec{k} \begin{vmatrix} 3 & 2 \\ 4 & -3 \end{vmatrix}$$

$$\vec{C} = \vec{i} [(2)(2) - (-3)(4)] - \vec{j} [(3)(2) - (4)(4)] + \vec{k} [(3)(-3) - (2)(2)]$$

$$\vec{C} = \vec{i} [4 + 12] - \vec{j} [6 - 16] + \vec{k} [-9 - 4]$$

$$\vec{C} = 16\vec{i} + 10\vec{j} - 13\vec{k}$$

Area del paralelogramo

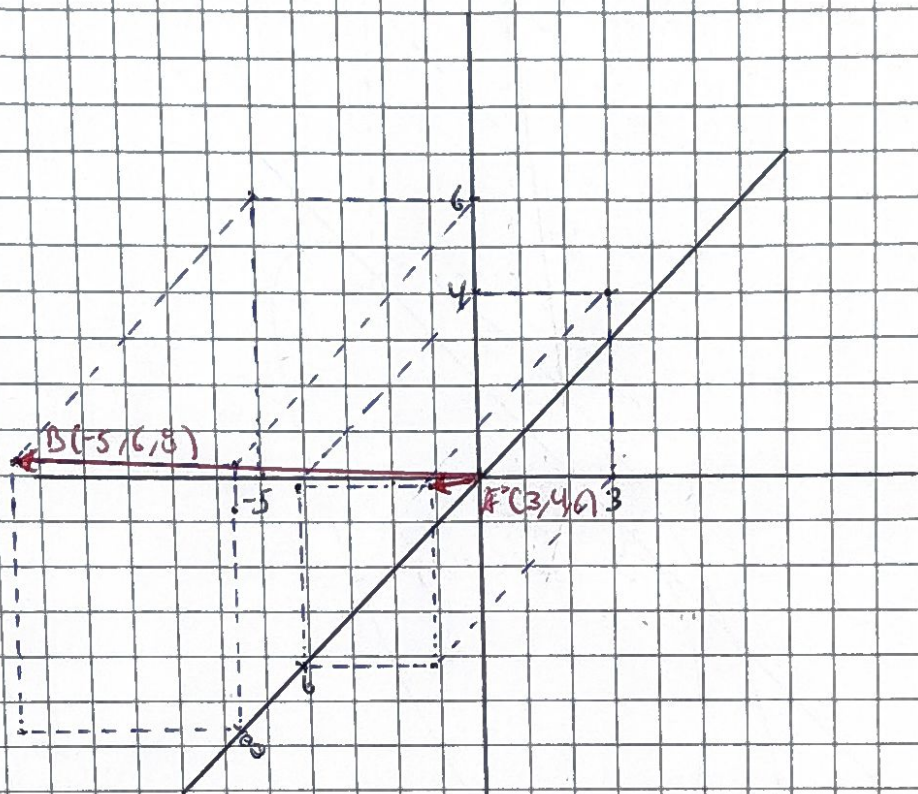
$$|\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin \theta$$

$$|\vec{A} \times \vec{B}| = \sqrt{29} \cdot \sqrt{29} \cdot \sin(61,13^\circ)$$

$$|\vec{A} \times \vec{B}| = 29 \cdot 0,8757$$

$$|\vec{A} \times \vec{B}| = 25,39$$

6. $\vec{A} = 3\vec{i} + 4\vec{j} + 6\vec{k}$
 $\vec{B} = -5\vec{i} + 6\vec{j} + 8\vec{k}$



Modulo de A.

$$|\vec{A}| = \sqrt{(3)^2 + (4)^2 + (6)^2}$$

$$|\vec{A}| = \sqrt{9 + 16 + 36}$$

$$|\vec{A}| = \sqrt{61}$$

$$|\vec{A}| = 7,81$$

Modulo de B

$$|\vec{B}| = \sqrt{(-5)^2 + (6)^2 + (8)^2}$$

$$|\vec{B}| = \sqrt{25 + 36 + 64}$$

$$|\vec{B}| = \sqrt{125}$$

$$|\vec{B}| = 11,18$$

Producto Escalar

$$\vec{A} \cdot \vec{B} = (A_x B_x) + (A_y B_y) + (A_z B_z)$$

$$\vec{A} \cdot \vec{B} = (3)(-5) + (4)(6) + (6)(8)$$

$$\vec{A} \cdot \vec{B} =$$

7. $\vec{A} = 2\vec{i} - 1\vec{j} + 2\vec{k}$
 $\vec{B} = -3\vec{i} + 6\vec{j} - 6\vec{k}$

